

Assignment 6

Machine	Temperature	Vibration	Failure
1	High	High	Yes
2	High	Low	Yes
3	Medium	High	Yes
4	Low	Low	No
5	Low	Medium	No
6	Medium	Low	No
7	High	Medium	Yes
8	Low	High	No

Calculate Total Entropy

Failure:

Yes = 4

No = 4

Total = 8

$$\begin{aligned}
 \text{Entropy (S)} &= - \sum_{i=1}^2 P_i \log_2 P_i \\
 &= - \frac{4}{8} \log_2 \left(\frac{4}{8} \right) - \frac{4}{8} \log_2 \left(\frac{4}{8} \right) \\
 &= 1
 \end{aligned}$$

Information gain for temperature

Temperature = High

Failure

Yes = 3

No = 0

$$\text{Entropy (High)} = - \frac{3}{3} \log_2 \left(\frac{3}{3} \right) = 0$$

Temperature = Medium

Yes = 1

No = 1

$$\text{Entropy (Medium)} = -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right)$$

$$= 1$$

Temperature = low

$$\text{No} = 3$$

$$\text{Entropy (low)} = 0$$

Weighted Entropy for Temperature
(S, temp)

$$\text{Entropy}_n = \frac{5}{8} \times 0 + \frac{2}{8} \times 1 + \frac{1}{8} \times 0$$

$$= 0.25$$

Information gain for temperature

$$\text{Gain (S, temp)} = \text{Entropy (S)} - \text{Entropy (S, temp)}$$

$$= 1 - 0.25$$

$$= 0.75$$

Information Gain for vibration

Vibration = High

$$\text{Yes} = 2$$

$$\text{No} = 1$$

$$\text{Entropy} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \left(\frac{1}{3}\right)$$

$$= 0.918$$

Vibration = Medium

$$\text{Yes} = 1$$

$$\text{No} = 1$$

$$\text{Entropy} = 1$$

Vibration = Low

$$\text{Yes} = 1$$

$$\text{No} = 2$$

$$\text{Entropy} = 0.918$$

Weighted Entropy

$$\begin{aligned}\text{Entropy}(S, \text{vibration}) &= \frac{5}{8} \times 0.918 + \frac{2}{8} \times 1 + \frac{3}{8} \times 0.918 \\ &= 0.9385\end{aligned}$$

Information gain

$$\begin{aligned}\text{Gain}(S, \text{vibration}) &= \text{Entropy}(S) - \text{Entropy}(S, \text{vibration}) \\ &= 1 - 0.9385 \\ &= 0.0615\end{aligned}$$

Root Node

Attribute	Information gain
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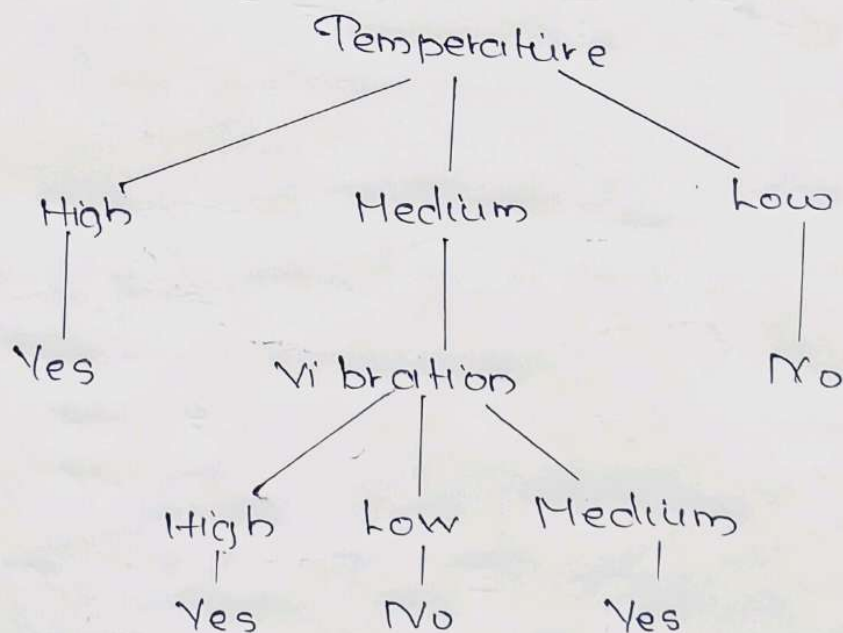
Temperature	0.75
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vibration	0.0615
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~~Gain~~ $\text{Gain}(S, \text{temp}) > \text{Gain}(S, \text{vibration})$

Root Node = Temperature

Decision Tree



Prediction (Temperature = High, Vibration = Medium)

Temperature = High

↓
Yes

Machine Failure = Yes